



Research article

Genome-wide analysis of plant-type II Ca^{2+} ATPases gene family from rice and Arabidopsis: Potential role in abiotic stresses

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ABSTRACT

The Plant Ca^{2+} ATPases are members of the P-type ATPase superfamily and play essential roles in pollen tube growth, vegetative development, inflorescence architecture, stomatal opening or closing as well as transport of Ca^{2+} , Mn^{2+} and Zn^{2+} . Their role in abiotic stress adaptation by activation of different signaling pathways is emerging. In Arabidopsis, the P-type Ca^{2+} ATPases can be classified in two distinct groups: type IIA (ECA) and type IIB (ACA). The availability of rice genome sequence allowed performing a genome-wide search for P-type Ca^{2+} ATPases proteins, and the comparison of the identified proteins with their homologs in Arabidopsis model plant. In the present study, we identified the P-type II Ca^{2+} ATPases from rice by analyzing their phylogenetic relationship, multiple alignment, cis-regulatory elements, protein domains, motifs and homology percentage. The phylogenetic analysis revealed that rice type IIA Ca^{2+} ATPases clustered with Arabidopsis type IIA Ca^{2+} ATPases and showed high sequence similarity within the group, whereas rice type IIB Ca^{2+} ATPases presented variable sequence similarities with Arabidopsis type IIB members. The protein homology modeling, identification of putative transmembrane domains and conserved motifs of rice P-type II Ca^{2+} ATPases provided information on their functions and structural architecture. The analysis of P-type II Ca^{2+} ATPases promoter regions in rice showed multiple stress-induced cis-acting elements. The expression profile analysis indicated vital roles of P-type II Ca^{2+} ATPases in stress signaling, plant development and abiotic stress responses. The comprehensive analysis and expression profiling provided a critical platform for functional characterization of P-type II Ca^{2+} ATPase genes that could be applied in engineering crop plants with modified calcium signaling and homeostatic pathways.

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1. Introduction

Ca^{2+} acts as a secondary messenger and is involved in regulation of growth, development, reproduction, circadian rhythms, immunity, redox status, hormone biosynthesis as well as response to biotic and abiotic stresses by activation of various signaling pathways [1]. Plant Ca^{2+} ATPases are members of P-type ATPase superfamily implicated in the restoration and maintenance of ion homeostasis by pumping calcium ions out of the cytosol [2]. Plant P-type Ca^{2+} ATPases have been divided into two groups, type IIA and IIB, the latter contains an N-terminal autoinhibitory domain that binds to calmodulin and activates the Ca^{2+} pump [3]. Type IIA

and IIB Ca^{2+} ATPases can be found in either or both plasma membrane and endomembrane systems [3]. This distribution is in contrast with animal cells, where type IIA and IIB Ca^{2+} ATPases are found exclusively in the inner membranes and plasma membrane (PM), respectively. In Arabidopsis, type IIA Ca^{2+} ATPases are extensively localized in the endoplasmic reticulum (ER) and Golgi complex [4]. Unlike the type IIA Ca^{2+} ATPases, the type IIB Ca^{2+} ATPases have been reported in various subcellular structures. For instance, At-ACA1 and At-ACA2 have been reported in chloroplast and ER [4], At-ACA4 and At-ACA11 were localized in vacuole [4,5], while At-ACA7 [6], At-ACA8 [4], At-ACA9 [4] are localized in PM, while At-ACA10 has been predicted to be plasma membrane localized [7]. Their variable localization in different cellular compartments could indicate the involvement in different Ca^{2+} mediated signaling pathways.

The P-type II Ca^{2+} ATPases are characterized by a high homology (50–80%) at amino acid level. The structures obtained from crystals of mammalian type IIA Ca^{2+} ATPase showed the presence of 10 transmembrane helices [8], while the hydrophobicity analysis in

Abbreviations: ACA, autoinhibited Ca^{2+} ATPase; ECA, ER-type Ca^{2+} ATPase; ER, endoplasmic reticulum; PCA1, *Physcomitrella* Ca^{2+} ATPase; PCD, programmed cell death; qRT-PCR, quantitative reverse transcription PCR; SCA1, Soybean Ca^{2+} ATPase; TM, transmembrane.

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the primary structure sometimes showed eight transmembrane domains in plant Ca^{2+} ATPases [3]. All P-type ATPases have characteristically nine conserved motifs [9]. The first three motifs are located in the small cytoplasmic loop between TM2 and TM3 and play a role in attaining the auto-inhibited state, as in the case of At-ACA8 [10]. Motifs 5–8 span between TMSs 4 and 5 and are involved in nucleotide binding and phosphorylation. Motif 5 may play a role in catalysis while motif 6 is thought to have a part in ATP binding [11]. Residues in motif 7 are regarded important for phosphate binding and phosphorylation of motif 5 [11]. The last three residues in motif 8 are also believed to be critical for phosphorylation of the enzymes [9]. Motif 9 has been stated to play a role in inducing conformational changes in transition state of the protein [11].

Plant Ca^{2+} ATPases are involved in many fundamental processes such as pollen tube growth [6], vegetative development, inflorescence architecture [7], stomatal opening or closing [12] and gibberellin signaling [13]. Type II A Ca^{2+} ATPases also play roles in Ca^{2+} , Mn^{2+} and Zn^{2+} transport. The previous reports on *At-ECA1* (Endomembrane type Calcium ATPase) demonstrated the role this gene played in providing tolerance to manganese toxicity [14], while *At-ECA3* was required in Mn^{2+} and Ca^{2+} homeostasis [15]. Based on their stress-induced expression levels, P-type II Ca^{2+} ATPases have also been predicted to act in abiotic and biotic stress adaptation [16]. The expression of many Ca^{2+} ATPases increased above their basal level upon exposure to various stresses, while others are specifically expressed only under stress conditions [3]. For instance, the *PCA1* relative mRNA level was up-regulated under dehydration, NaCl and abscisic acid treatments [16]. The expression of a type IIA Ca^{2+} ATPase from tomato [17] and tobacco [18] was induced by NaCl. The plasma membrane Ca^{2+} ATPase *SCA1* from soybean is highly and rapidly induced by NaCl stress but not by osmotic stress [19]. The Arabidopsis *ACA4* gene is also up-regulated by NaCl treatment [3]. The publically available microarray data from AtGenExpress (<http://jsp.weigeltworld.org/expviz/expviz.jsp>) revealed distinct basal and induced expression patterns of different Ca^{2+} ATPases. The *ACA* (Auto inhibited type calcium ATPase) gene expression is significantly regulated by stress treatments, especially by NaCl treatment. The induced expression of Ca^{2+} ATPase genes suggests that the Ca^{2+} pumps might be part of Ca^{2+} -dependent signal transduction pathway linked to abiotic stresses. In Arabidopsis, the *ACA8* and *ACA9* expression is stimulated by abscisic acid and thus, an indirect evidence for the role of plant Ca^{2+} ATPases in stress signaling based on the ABA-responsive expression was provided [20]. The calcium signals not only regulate plant development and abiotic stresses, but also protect the plant from pathogens by activating salicylic acid (SA)-mediated programmed cell death (PCD) pathways [5].

The present study, however, focused on the exploration of the complete set of Ca^{2+} ATPases genes present in the two sequenced model plant genomes, rice and Arabidopsis, by bioinformatic approaches. The Ca^{2+} ATPase genes from rice and Arabidopsis were compared, based on the analysis of phylogenetic relationships, multiple alignment, important cis-regulatory elements, protein domains, different motifs and homology percentage. The gene expression patterns of 15 Ca^{2+} ATPase genes in rice under various stress conditions and developmental stages were analyzed and validated by qRT-PCR analysis under stress conditions. The results presented here shed light on several key aspects of Ca^{2+} ATPase gene family phylogeny and the distribution of conserved motifs and provide insights into the structure and composition of Ca^{2+} ATPase gene families in rice and Arabidopsis. The classification of the rice Ca^{2+} ATPases provides a backbone for more detailed analysis of type IIA and IIB Ca^{2+} ATPases in rice.

2. Results

2.1. Predominant features of plant-type II Ca^{2+} ATPases in rice and Arabidopsis

In rice genome annotation database, a keyword-based search by the name “Plasma membrane Calcium ATPase” fetched 21 hits. A sequence-based search added 10 more genes in the study. From the 31 retrieved Ca^{2+} ATPases, S.M.A.R.T. and Pfam software could identify the Ca^{2+} binding domain in only 15 genes, while the other 16 genes in which the characteristic domains could not be found were excluded from further analysis. The P-type II Ca^{2+} ATPases identified in rice and Arabidopsis genome database and their features are mentioned in (Tables 1 and 2). Out of the 15 rice Ca^{2+} ATPases, three (LOC_Os03g17310, LOC_Os03g52090 and LOC_Os05g02940) were defined as P-type IIA Ca^{2+} ATPases, while the remaining 12 sequences were classified as P-type IIB Ca^{2+} ATPases (Table 1). The Ca^{2+} ATPase family genes possess a large number of introns in their sequences with LOC_Os03g52090 having a maximum of 36 introns. Amongst the type IIA Ca^{2+} ATPases, only LOC_Os05g02940 had two alternate spliced forms. The type IIB family exhibited very little sequence variation in terms of number of amino acids but presented high variation in terms of number of introns (Table 1). Within the type IIB members, only LOC_Os04g51610 and LOC_Os08g40530 presented alternate spliced forms. The four members of Arabidopsis type IIA Ca^{2+} ATPases (*ECA1*–*4*) also presented low variation in protein length, high variation in number of introns and did not possess alternate spliced forms (Table 2). *At-ECA1* and *At-ECA3* were localized in ER and Golgi apparatus, while sub-cellular location of *At-ECA2* and *At-ECA4* was not reported. Eleven members of type IIB Ca^{2+} -ATPases (*ACA* pumps) were identified in Arabidopsis (Table 2), out of which only two (*At-ACA1* and *At-ACA8*) had alternate spliced forms. While *At-ACA7*, *At-ACA8*, *At-ACA9* have been shown to be localized on PM, *At-ACA10*, due to its high sequence homology to these proteins, has been predicted to be PM-localized. All these four proteins have an extended N-terminus and are slightly larger (except for *At-ACA7*) than other plant type IIB Ca^{2+} ATPase. Other members of type IIB Ca^{2+} ATPase showed multiple location; *At-ACA4*, *At-ACA11* were localized in vacuole, whereas *At-ACA1* and *At-ACA2* were localized in chloroplast and ER respectively (Table 2). The localization predicted by the databases was cross-checked with the available literature.

2.2. Classification and domain analysis of rice type II Ca^{2+} ATPases gene family

The results of conserved domains analysis of rice type II Ca^{2+} ATPase proteins were identified, based on the results of Pfam and SMART has been shown in (Fig. 1). The five domains characteristic for Ca^{2+} ATPase proteins, namely N-terminal autoinhibitory domain, N-terminus cations transporter/ATPase domain, E1-E2 ATPase domain, haloacid dehalogenase-like hydrolase and C-terminus cations transporter/ATPase were found to be conserved among most of the type II Ca^{2+} -ATPase proteins. The criterion for classification of P-type IIB Ca^{2+} ATPases from Arabidopsis was based on presence or absence of N-terminal autoinhibitory domain. This classification criterion did not suffice for two rice Ca^{2+} ATPase genes (LOC_Os02g08010 and LOC_Os08g40530) that exhibited high sequence similarity with other P-type IIB Ca^{2+} ATPases but did not possess this domain (Fig. 1). Therefore, in our classification of rice IIB Ca^{2+} ATPase proteins, we have excluded the criteria of presence of N terminal auto-inhibitory domain as a primary requisite, but have carried out the classification on the basis of sequence and phylogenetic similarity amongst the member proteins.

Table 1
Plant-type II Ca²⁺ ATPases in the rice genome.

SL. No.	TIGR locus ID	Gene name	KOME ACC	Size (bp)	Size (AA)	Introns	Alt. splicing
P-type Ca ²⁺ ATPase IIB							
1	LOC_Os01g71240	Os-ACA1	AK070064	3132	1044	6	
2	LOC_Os02g08010	Os-ACA2	AK242506	1776	592	16	
3	LOC_Os02g08018	Os-ACA3	N/A	978	326	9	
4	LOC_Os03g10640	Os-ACA4	AK121250	3060	1020	6	
6	LOC_Os03g42020	Os-ACA5	AK065088	3102	1034	6	
8	LOC_Os04g51610	Os-ACA6	AK120057	3267	1089	33	2
10	LOC_Os05g41580	Os-ACA7	N/A	3222	1074	8	
11	LOC_Os08g40530	Os-ACA8	AK058787	1818	606	18	2
12	LOC_Os10g28240	Os-ACA9	AK109037	3108	1036	0	
13	LOC_Os11g04460	Os-ACA10	AK242630	3054	1018	6	
14	LOC_Os12g04220	Os-ACA11	N/A	3120	1040	6	
15	LOC_Os12g39660	Os-ACA12	AK066259	3063	1021	6	
P-type Ca ²⁺ ATPase IIA							
5	LOC_Os03g17310	Os-ECA1	AK105840	3189	1063	7	
7	LOC_Os03g52090	Os-ECA2	AK073220	3654	1218	36	
9	LOC_Os05g02940	Os-ECA3	AK062874	1122	374	3	2

2.3. Sequence divergence and phylogenetic analysis

The percentage of identity and divergence in the amino acid sequence of all selected P-type II Ca²⁺ATPase was identified using MEGALIGN program (Supplementary Fig. 1). The results of this analysis showed that At-ECA4 was closely related to At-ECA1 sharing 97.3% similarity, while 85% and 77.7% similarity was evident for LOC_Os05g02940 and LOC_Os03g17310, respectively. At-ECA4 also shared more than 49.4% similarity with other members of the Arabidopsis type IIA group (Supplementary Fig. 1). Similarly, type IIB Ca²⁺ATPases from Arabidopsis and rice were also compared. Among these, LOC_Os11g04460 was closely related to LOC_Os12g04220 sharing 98.2% similarity while it showed more divergence with AT5G53010 (33.4%) although they belong to the same group. At-ACA1 gene showed more than 65% similarity with At-ACA2 (77.9%), At-ACA7 (77.3%), LOC_Os12g39660 (76.7%), LOC_Os03g10640 (76.6%) and LOC_Os03g42020 (67.5%). At-ACA4 presented 87.6% similarity with At-ACA11 and 63.5% similarity with At-ACA2 while it shared more than 45% similarity with the members of type IIB Ca²⁺ATPases in rice (Supplementary Fig. 1). At-ACA8 had homology to other three type IIA Ca²⁺ATPase genes: At-ACA7 (49.5%), At-ACA9 (69.9%) and At-ACA10 (72.9%).

In order to determine the evolutionary relatedness among the members of P-type II Ca²⁺ATPases from rice and Arabidopsis, a phylogenetic tree was constructed by ClustalX software using the

deduced amino acid sequences (Fig. 2). The phylogenetic analysis revealed that P-type II Ca²⁺ATPases formed two distinct groups, namely type IIA and type IIB (Fig. 2, shaded in green and yellow, respectively). Members of the same group seemed to have high degree of evolutionary relatedness. The investigation for the evolution of P-type II Ca²⁺ATPases between the two plant species revealed that the members of respective groups from different plant species fell in same clades with high degree of similarity (Fig. 2). This pattern suggests the common ancestry of P-type II Ca²⁺ATPases in the two distinct plant species. In the phylogram, segmentally (LOC_Os11g04460 and LOC_Os12g04220) and tandemly duplicated genes (LOC_Os02g08010 and LOC_Os02g08018) were neighbors of each other and have been shown in gray and pink boxes respectively (Fig. 2). The genes from the same species with high sequence similarity and 1000 bootstrap value were regarded as homologs of each other and such gene pairs have been shown in blue shaded boxes while the ortholog gene pairs are shown in red boxes (Fig. 2).

2.4. Genome organization of plant-type II Ca²⁺-ATPases

The chromosomal localization analysis of rice type II Ca²⁺ATPases revealed variable distribution of the genes in all chromosomes except for chromosome 6, 7 and 9 (Fig. 3(A)). While a maximum of four genes were located on chromosome 3, the

Table 2
Plant-type II Ca²⁺ ATPases in the Arabidopsis genome.

SL. No.	TAIR LOC	Gene name	EST	Size (bp)	Size (AA)	Introns	Alt. splicing	Cellular location
P-type Ca ²⁺ ATPase IIA								
1	AT1G07810	At-ECA1	42	3186	1061	8		ER
2	AT4G00900	At-ECA2	10	3165	1054	5		
3	AT1G10130	At-ECA3	22	2997	998	33		Golgi
4	AT1G07670	At-ECA4	28	3186	1061	8		
P-type Ca ²⁺ ATPase IIB								
5	AT1G27770	At-ACA1	33	3063	1020	6	2	Chloroplast
6	AT4G37640	At-ACA2	50	3045	1014	6		ER
7	AT2G41560	At-ACA4	48	3093	1030	6		Vacuole
8	AT2G22950	At-ACA7	N/A	3048	1015	6		Plasma membrane
9	AT5G57110	At-ACA8	75	3225	1074	34	2	Plasma membrane
10	AT3G21180	At-ACA9	4	3261	1086	31		Plasma membrane
11	AT4G29900	At-ACA10	47	3210	1069	34		Plasma membrane
12	AT3G57330	At-ACA11	23	3078	1025	6		Vacuole
13	AT3G22910		4	3054	1017			
14	AT3G63380		10	3102	1033			
15	AT5G53010		3	3150	1049	32		

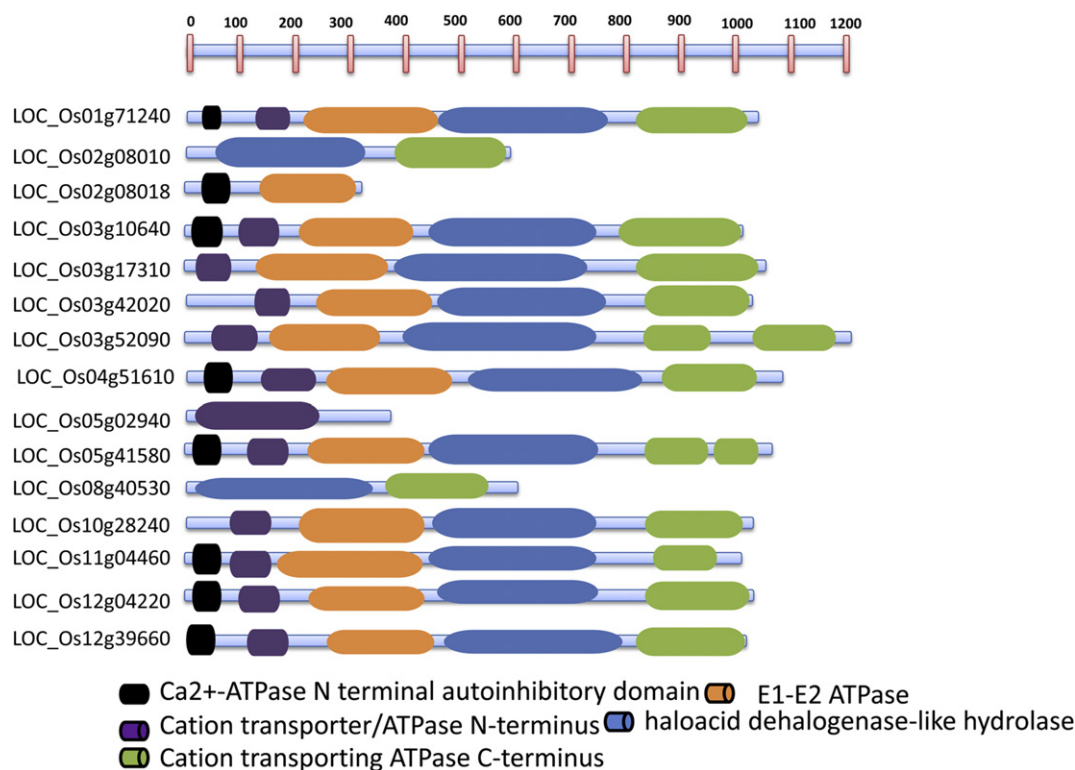


Fig. 1. Representation of domain structures in plant-type II Ca²⁺ ATPase proteins in *Oryza sativa*. The type II Ca²⁺ ATPases were classified into two groups: type IIA (lacking N-terminal autoinhibitory domain) and type IIB (contained N-terminal autoinhibitory domain). Other domains found in the Ca²⁺ ATPase family, namely: cation transporter/ATPase N-terminus, E1-E2 ATPase, haloacid dehalogenase-like hydrolase and cation transporter/ATPase C-terminus have also been depicted.

chromosomes 2, 5 and 12 had two type-II Ca²⁺ ATPase genes each (Fig. 3(A)). A pair of genes (LOC_Os11g04460 and LOC_Os12g04220) on chromosome 11 and 12 was found to be segmentally duplicated (marked in Fig. 3(A) by a red dotted arc). On the other hand, another

pair of genes (LOC_Os02g08010 and LOC_Os02g08018) was marked as tandemly duplicated, based on the criterion of separation by less than five consecutive genes (Fig. 3(A), bracketed together in green color).

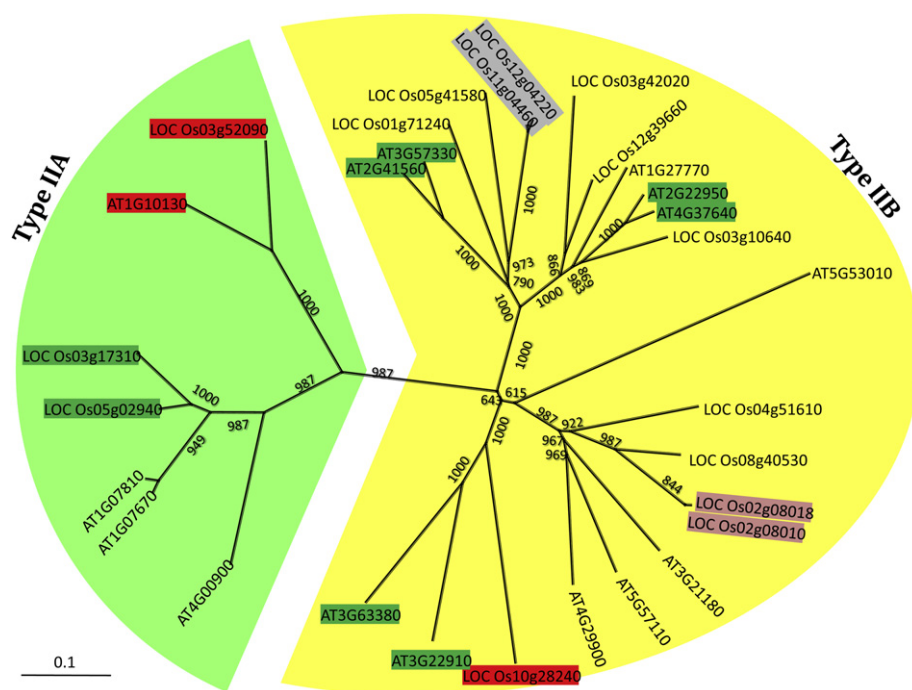


Fig. 2. Phylogenetic relationship among plant-type II Ca²⁺ ATPases in rice and Arabidopsis. Tree was constructed using neighbor-joining method with 1000 bootstrap replicates. Green shade represents p-type IIA and yellow represents type IIB Ca²⁺ ATPases. Segmentally and tandemly duplicated genes were shown in gray and pink boxes respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

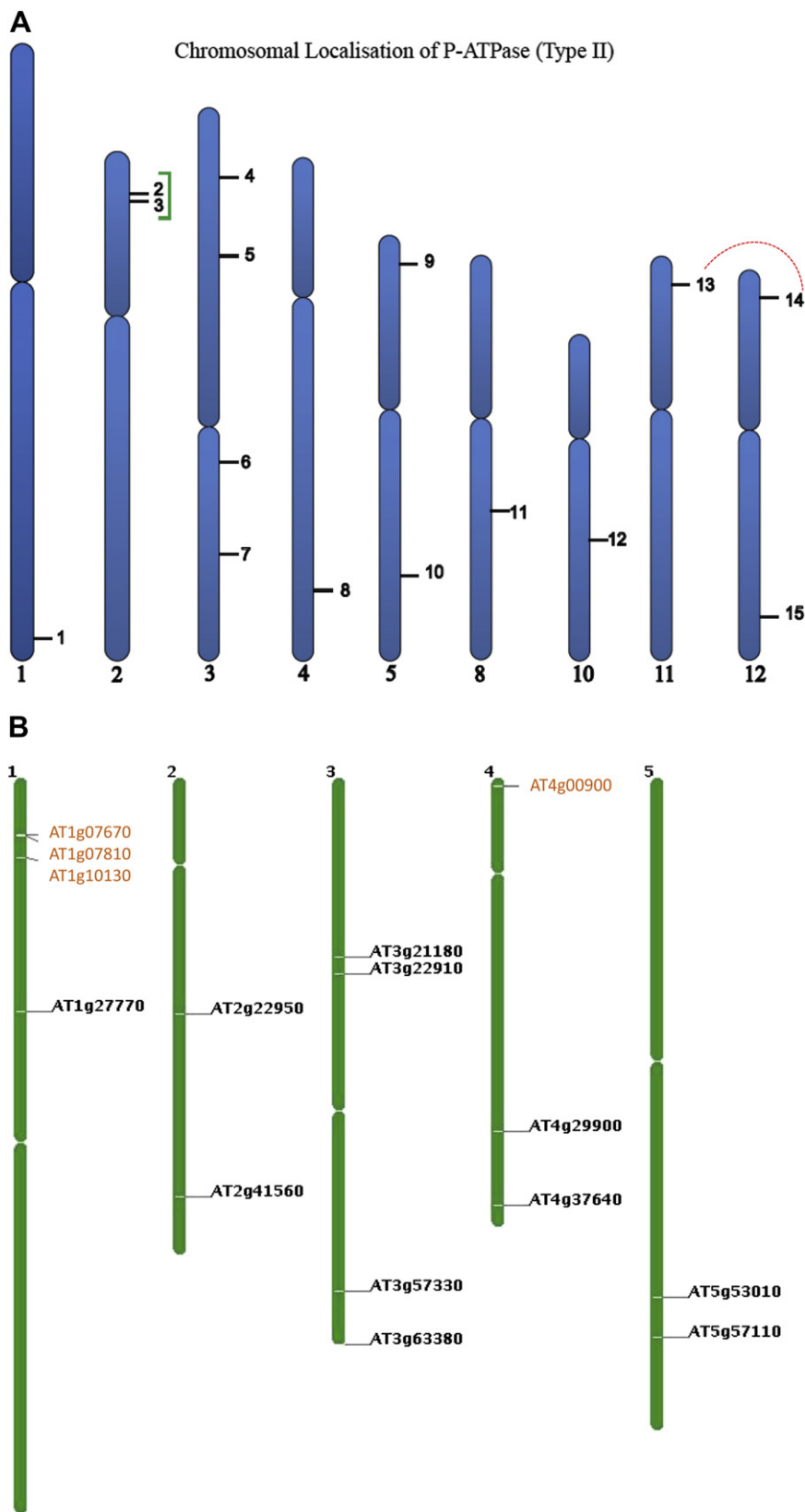


Fig. 3. Genome organization of rice and Arabidopsis plant-type II Ca^{2+} ATPase genes. (A) Chromosomal localization of rice p-type II Ca^{2+} ATPase genes on the 12 chromosomes of rice. Genes from different type II classes have been mapped by their position on the 12 chromosomes. Segmental duplications have been marked by red arc while the tandemly duplicated genes are joined with vertical green lines. No type II gene was localized to chromosome 6, 7 and 9. Respective chromosome numbers are mentioned at the bottom. (B) Genome organization of Arabidopsis type II Ca^{2+} ATPase genes. The figure depicts the arrangement of type II genes on the five Arabidopsis chromosomes. Respective chromosome numbers are mentioned at the top. Red dotted lines depict duplicated segments of the genome. Tandemly duplicated genes are joined with black green lines. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Similarly, the genome organization of Arabidopsis P-type II Ca^{2+} ATPases family was also analyzed. Type IIA Ca^{2+} ATPases from Arabidopsis are depicted in orange color (Fig. 3(B)). Among them AT1G07810, AT1G10130 and AT1G07670 were localized to chromosome 1 and only one member (AT4G00900) on chromosome 4. Most of the type IIB Ca^{2+} ATPases genes (AT3G21180, AT3G22910, AT3G57330 and AT3G63380) were located on chromosome 3. Chromosome 2, 4 and 5 each contained two members of type IIB Ca^{2+} ATPase genes, whereas only one type IIB member (AT1G27770) is localized on chromosome 1 (Fig. 3(B)). Unlike rice, no gene duplication event was observed in Arabidopsis.

2.5. Plant-type II Ca^{2+} ATPases: motif to structural analysis

In order to detect the conserved domains and motifs among the P-type II Ca^{2+} ATPases, multiple sequence alignments were carried out. The N-terminal autoinhibitory calmodulin-binding domain (CMBD) was observed in most of the P-type IIB Ca^{2+} ATPases, with the exception of LOC_Os02g08010 and LOC_Os08g40530 (Fig. 4), whereas none of the P-type IIA proteins possessed this domain. Other predicted P-type II Ca^{2+} ATPases transmembrane domains, common to all higher organisms, were also observed (Fig. 4, designated as M1–M10). The inter-transmembrane and the non-conserved regions between M1 and 2, M4 and 5 and M6 to M10 have not been depicted in Fig. 4 due to the size constraints. It was noted that LOC_Os02g08010, LOC_Os05g02940 and LOC_Os08g40530 lacked the transmembrane domains M1–M5, while LOC_Os02g08018 showed the presence of transmembrane domains M2–M4 only.

The nine characteristic motifs common in all P-type ATPases were also evidenced (Fig. 4) (shown in red box). The Proline (P) residue in motif 1 was found to be replaced with Valine (V) in some cases, while the G (Glycine) and D (aspartate) residues were highly conserved in motif 1 of all P-type II Ca^{2+} ATPases. Of all the motifs, motif 2 (PAD) and motif 5 (DKTGTLT) were found to be best conserved without any exception among all the members investigated in the present study (Fig. 4). The motif 9 (VAMTGDG-ANDAPALKKADIGIAM, the hinge motif), was not well conserved in most of the members analyzed, although the internal TGDG sub-motif was highly conserved. Other residues within the hinge motif exhibited different degrees of conservation. Apart from these, we have identified two new motifs (TAKAIALECG and GTA-VAKE) (shown in black box, Fig. 4) flanking motif 9 on its either sides.

The search for the structural homologs was carried out using representative family members from rice (LOC_Os04g51610) and Arabidopsis (AT5G57110). Both of these genes are members of P-type IIB Ca^{2+} ATPases, while the template obtained from ESYPred3D (Web Server 1.0.) was 3TLM (PDB ID for Crystal Structure of Endoplasmic Reticulum Ca^{2+} ATPase (SERCA) from Bovine Muscle), which is P-type IIA Ca^{2+} ATPase. The rice and Arabidopsis P-type II Ca^{2+} ATPases showed 30–40% sequence similarity with the template.

The 3D modeled protein structures of P-type II Ca^{2+} ATPases from rice (LOC_Os04g51610) and Arabidopsis (AT5G57110), as observed using PyMol software, are shown in Fig. 5(A) and (B). Both the structure of rice and Arabidopsis P-type IIB Ca^{2+} ATPases revealed a transmembrane domain with 10 α -helices. Most of the α -helices were present in transmembrane region whereas the β sheets were on cytosolic region. The residues of the N-terminus Ca^{2+} -binding domain for both rice and Arabidopsis Ca^{2+} -ATPase structures have been evidenced in Fig. 5(C) and (D), respectively. In the case of rice, Tyr (141), Lys (152), Thr (157) and Val (204) formed coordinate bind with Ca^{2+} while in Arabidopsis Tyr (136; 173), Lys (152) and His (190) were involved in Ca^{2+} binding.

2.6. Expression profiling of plant-type II Ca^{2+} ATPases in rice and Arabidopsis

To check the expression profiling of various P-type II Ca^{2+} ATPases, the Affymetrix probes IDs were extracted from Rice Oligonucleotide Array Database. The Affymetrix ID of LOC_Os02g08018 and LOC_Os03g52090 were not found in the database; therefore these could not be included in the expression analysis. The rice P-type II Ca^{2+} ATPases were highly expressed in different developmental stages as well as in several stresses (Fig. 6(A)). LOC_Os01g71240 and LOC_Os03g17310 showed inflorescence specific variation in the expression pattern while in other stages, their expression pattern remained unaffected. Expression profiling of LOC_Os10g28240 illustrated that its expression was highly variable in various developmental stages: while in the inflorescence it is highly expressed except P5 stage, very low expression was evidenced in the shoot apical meristem (SAM). In case of seed stages, its expression decreased on a regular interval as the maturation stage proceeded. LOC_Os04g51610 showed a reduced expression in root, while in other stages (mature leaf, SAM, inflorescence and seed) its expression value was very high.

In order to test the Ca^{2+} ATPases expression profiles during various abiotic stresses, 7-day old seedlings were used for non-treated control and compared with drought, salt and cold stresses. LOC_Os01g71240, LOC_Os12g39660, LOC_Os03g17310 and LOC_Os05g02940 showed few or no changes in their expression patterns with all the used stresses. Other genes presented differential expression when submitted to these stresses. A handful of genes showed either single stress-specific transcript variation or in a combination of two stresses. LOC_Os08g40530 showed a drought specific expression variation, while the LOC_Os02g08010 expression was down-regulated under cold and salt stress. LOC_Os04g51610 showed drought- and salt-specific expression. Expression profiling of tandemly duplicated genes (LOC_Os02g08010 and LOC_Os02g08018), generated by comparing their expression values, and showed minute variations in the expression profile (Supplementary Fig. 2).

Genevestigator was used for hierarchical clustering of genes in a pattern, according to their expression in *Arabidopsis thaliana*. From a cluster of 13 genes, 10 were found to be differentially regulated under various abiotic stresses (Fig. 6(B)). AT3G21180, AT2G22950 and AT1G10130 presented no variation in their expression in any abiotic stresses, while the rest of the genes showed differential expression under various stresses. AT5G57110 showed very high expression under cold stress, while AT2G41560 showed considerably low expression in case of salt stress. In general, the gene family was more responsive to late cold stress, early drought, early heat and late salt stress treatments, while the response to oxidative stress was not very encouraging.

2.7. Validation of expression profile by quantitative real-time PCR

To validate the rice array database generated expression profile of the rice genes, drought, salt and cold treatments were chosen as the representative stresses for four P-type II Ca^{2+} ATPase. While comparing the relative transcript abundance (from qRT-PCR) with the expression profile generated from microarray data, a similar expression pattern was found for all the candidate genes (Fig. 7) except in the case of LOC_Os03g42020, where the expression pattern under drought stress differed from database generated values. In qRT-PCR, LOC_Os03g42020 exhibited highly up-regulated expression under salt and drought conditions, while the microarray data showed down-regulation in comparison with the control samples.

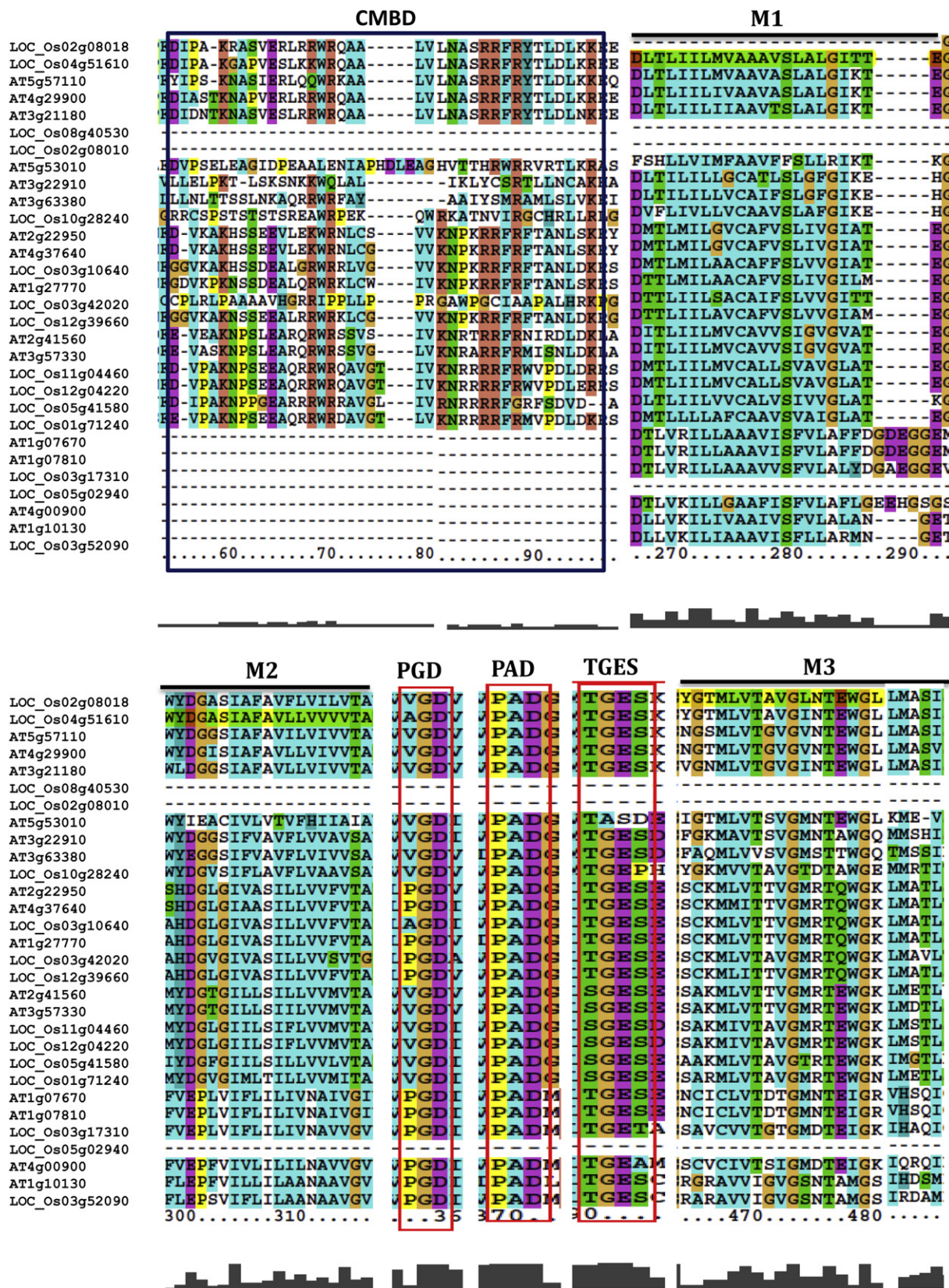


Fig. 4. Multiple amino acid sequence alignment of plant-type II Ca^{2+} ATPases from rice and Arabidopsis. Deduced amino acid sequences were aligned by the ClustalX software programme. The figure shows the conserved motifs (red boxes), the new identified motifs (black boxes) the predicted transmembrane domains (designated as M1–M10) (underlined by black lines on the top) and the putative CMBD (calmodulin binding domain) of P-type II Ca^{2+} ATPases (shown in blue box). The inter-transmembrane regions (with no prominent conservation) have been omitted from the alignment. The position of transmembrane was determined by using <http://www.membranetransport.org/>. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

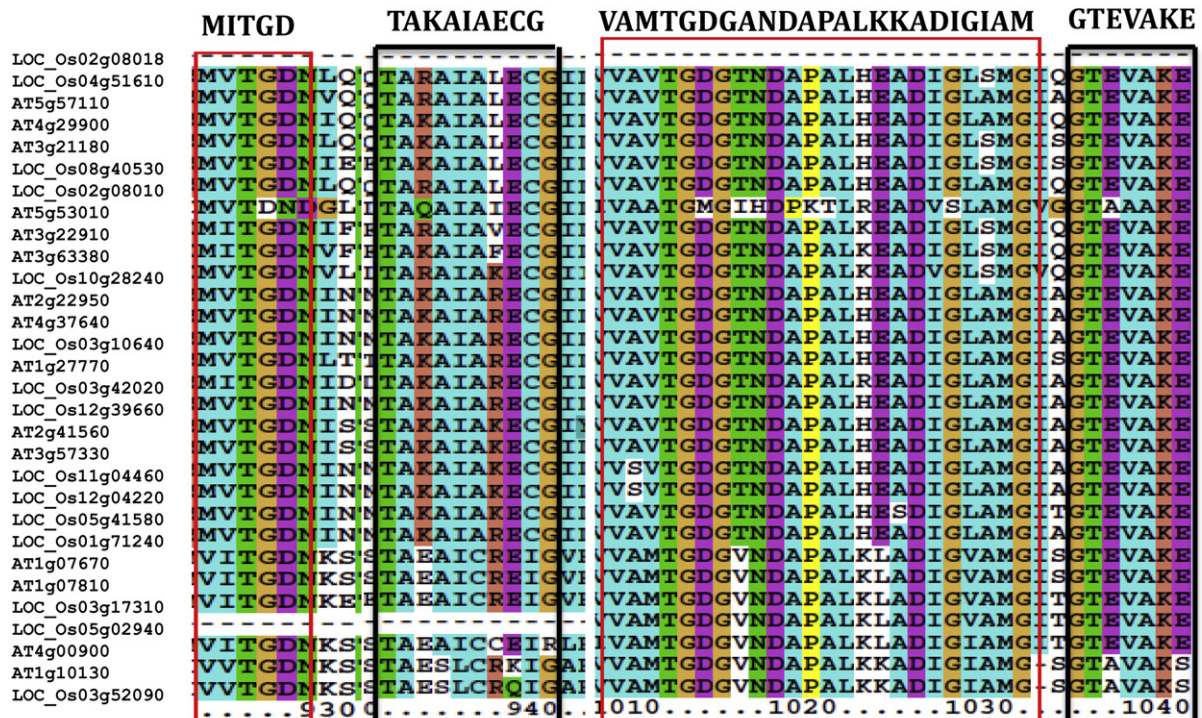
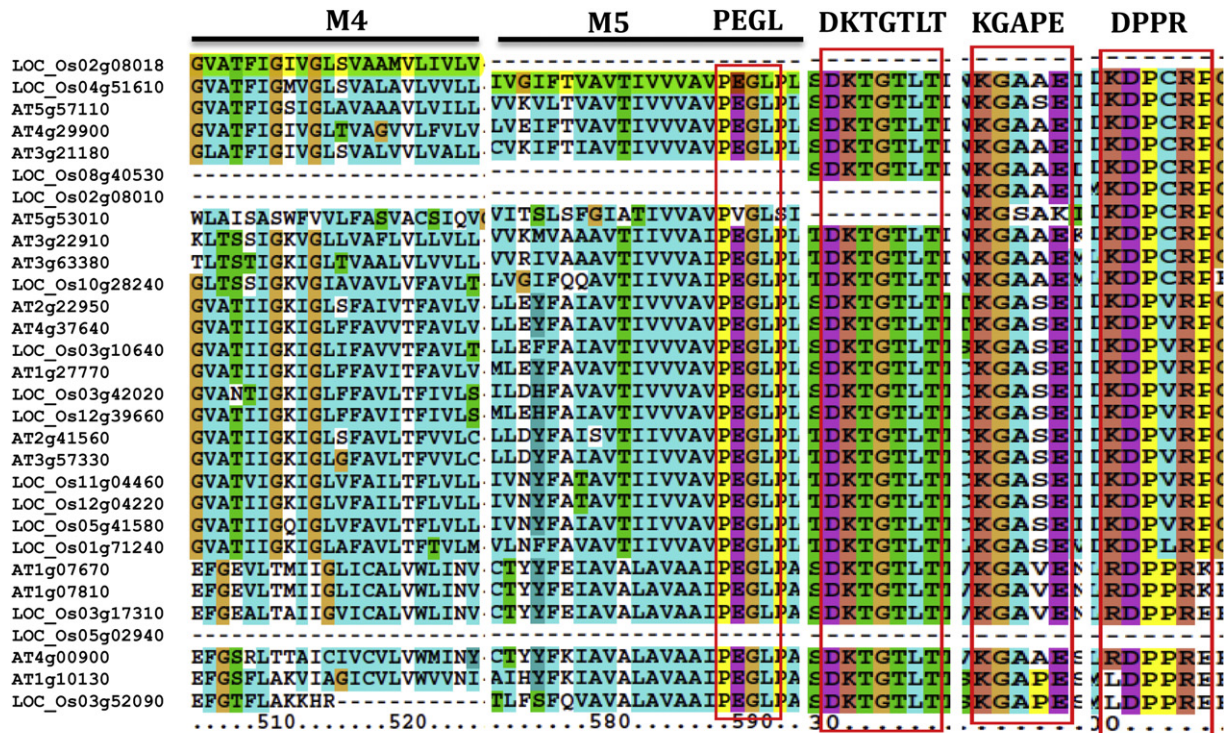


Fig. 4. (continued).

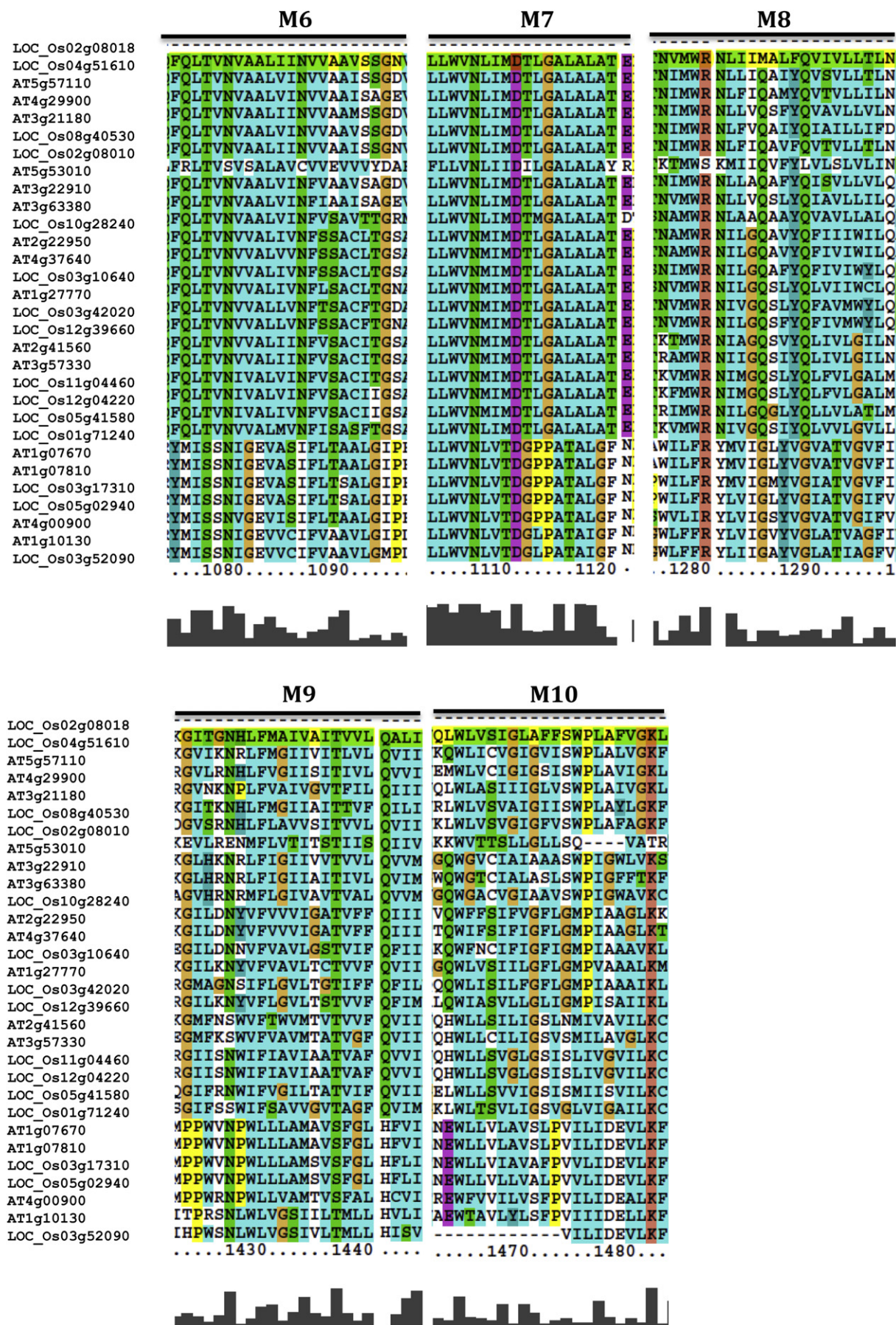


Fig. 4. (continued).

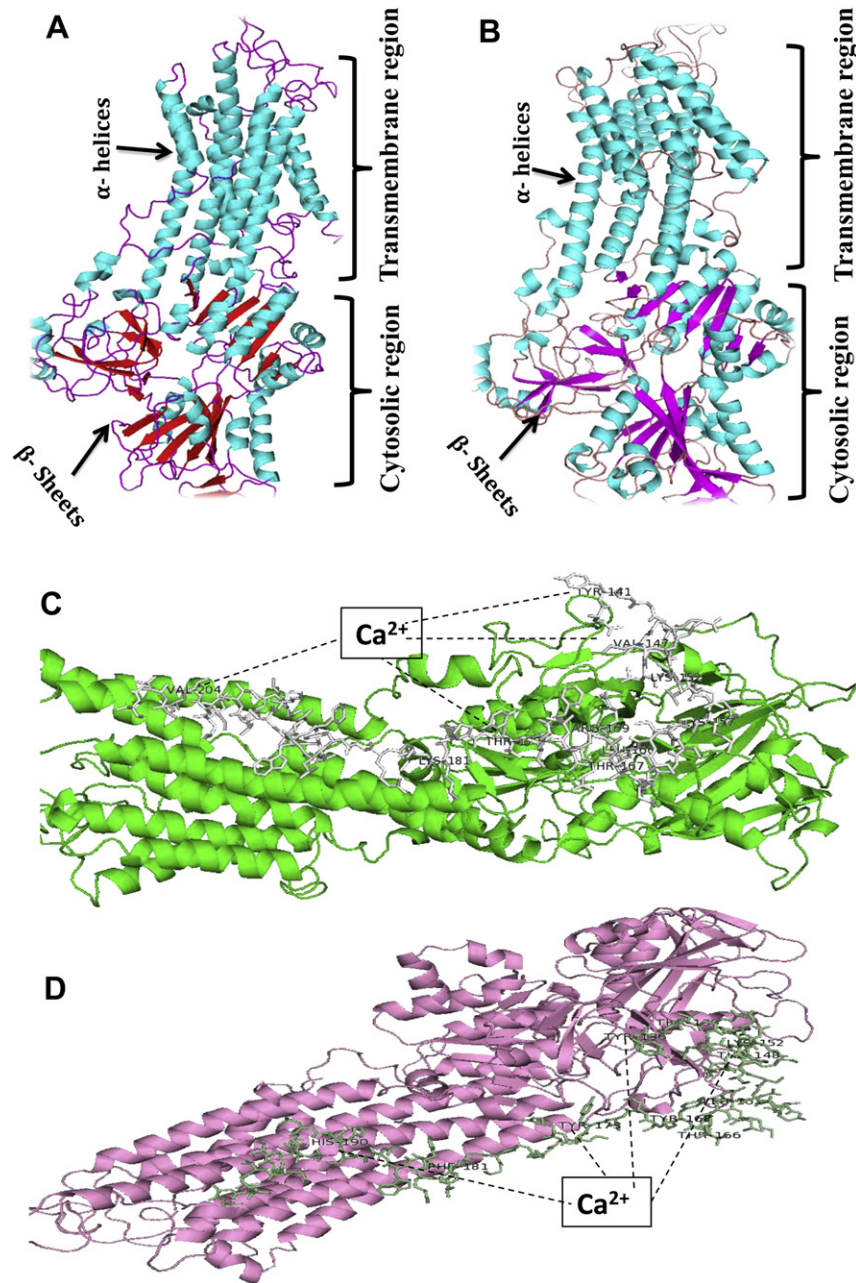


Fig. 5. Homology based model of representative plant-type IIB Ca^{2+} ATPase genes. (A) Rice, (B) Arabidopsis (B); the N-terminal Ca^{2+} -transporting ATPase domain in rice (C); N-terminal Ca^{2+} -transporting ATPase domain in Arabidopsis (D). The models were constructed using ESYPred3D Web Server 1.0 while viewed and edited in PyMol. Amino acid residues of the N-terminal Ca^{2+} binding sites have been marked to show possible coordination with the Ca^{2+} ion.

2.8. Multiple environmental and hormone-related cis-elements predicted in plant-type II Ca^{2+} ATPases promoter sequences in rice

Using PLACE and PLANTCARE databases, we analyzed the promoters of P-type II Ca^{2+} ATPases in rice and predicted their key cis-acting elements. The data revealed the presence of multiple stress-induced cis-acting elements in P-type II Ca^{2+} ATPases promoter regions. A detailed annotation including names, element core sequences and functions have been mentioned (Table 3). The common cis-elements found in all the promoters belonged to the MYB and the MYC family, known to be responsive toward drought, ABA and cold signals. Except for LOC_Os03g52090, LOC_Os02g08010 and LOC_Os02g08018, all sequences harbored pathogenesis and salt stress related cis-acting element

GT1GMSCAM4. While three type IIB Ca^{2+} ATPase genes possessed the GAREAT element, it was found to be absent in all type IIA Ca^{2+} ATPase promoters. The WBOXATNPR1 and light-responsive elements such as I-BOX, GT1 and GATABOX were also evidenced. Calcium responsive cis-element ABRERATCAL was also discovered in 50% of the promoters.

3. Discussion

Ca^{2+} ATPases are ubiquitous enzymes found in various organisms ranging from prokaryotes to mammals. In this report, we present a detailed analysis of P-type II Ca^{2+} ATPases encoded within rice and Arabidopsis genomes. The Arabidopsis genome contains four P-types IIA Ca^{2+} ATPases (ECA1–ECA4) whereas three

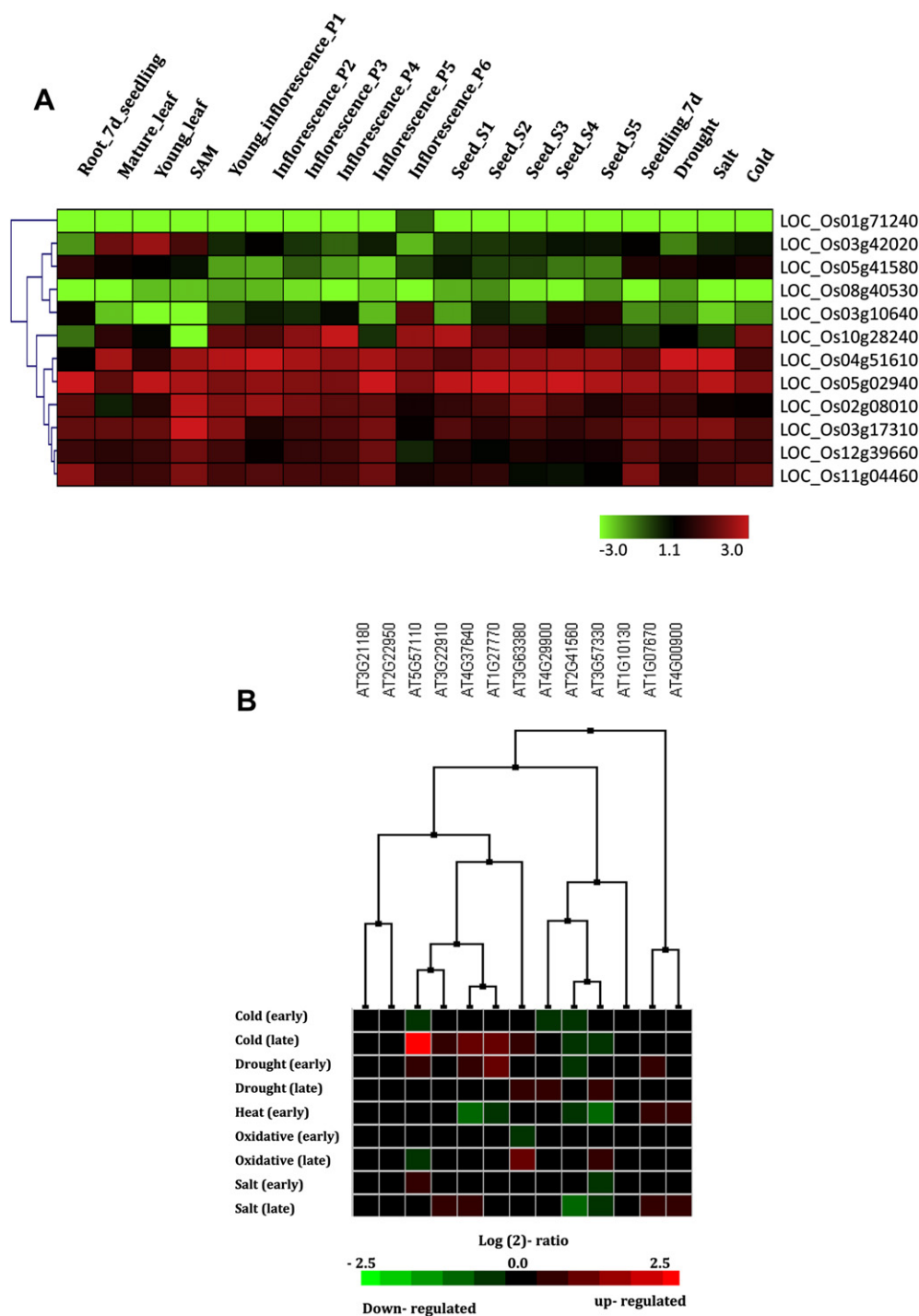


Fig. 6. Expression profile of the rice and Arabidopsis plant-type II Ca^{2+} ATPase genes in various developmental stages and stress conditions depicted as a heat map. (A) Expression profiles of rice p-type II Ca^{2+} ATPase genes. Reproductive development comprising six stages of panicle [P1 (0–3 cm), P2 (3–5 cm), P3 (5–10 cm), P4 (10–15 cm), P5 (15–22 cm), and P6 (22–30 cm)] and five stages of seed [S1 (0–2 DAP), S2 (3–4 DAP), S3 (4–10 DAP), S4 (11–20 DAP) and S5 (21–29 DAP)]. Clustering of the expression profile was done with log transformed average values taking mature leaf as base line. Three experimental stress conditions are denoted as drought stress, salt stress and cold stress. Different stresses were compared taking seven day old unstressed seedling. A gene is considered differentially expressed if it is up- or down-regulated by at least two-folds, with respect to the three vegetative controls (mature leaf, root and 7-day old seedling) and with respect to the seven day old unstressed seedling in case of abiotic stress. The color scale at the bottom of heat map is given in log 2 intensity value. (B) Expression profiles of Arabidopsis p-type II Ca^{2+} -ATPase genes. Hierarchical clustering was carried out using average linkage clustering rule. The color scale provided with heat map is given in log 2 intensity value.

members of this family were found in the rice genome, all related to the well-characterized SERCA Ca^{2+} ATPases in mammals. As for P-type IIB Ca^{2+} ATPases, which are related to the mammalian plasma membrane calmodulin-stimulated Ca^{2+} ATPases, 12 and 11 members were identified in rice and Arabidopsis, respectively. This is the

first report that presents the rice P-type IIB Ca^{2+} ATPase family members. Understanding the functional properties, expression patterns, and localization of each of these proteins is important for understanding their contribution to ion homeostasis and signaling in plants.

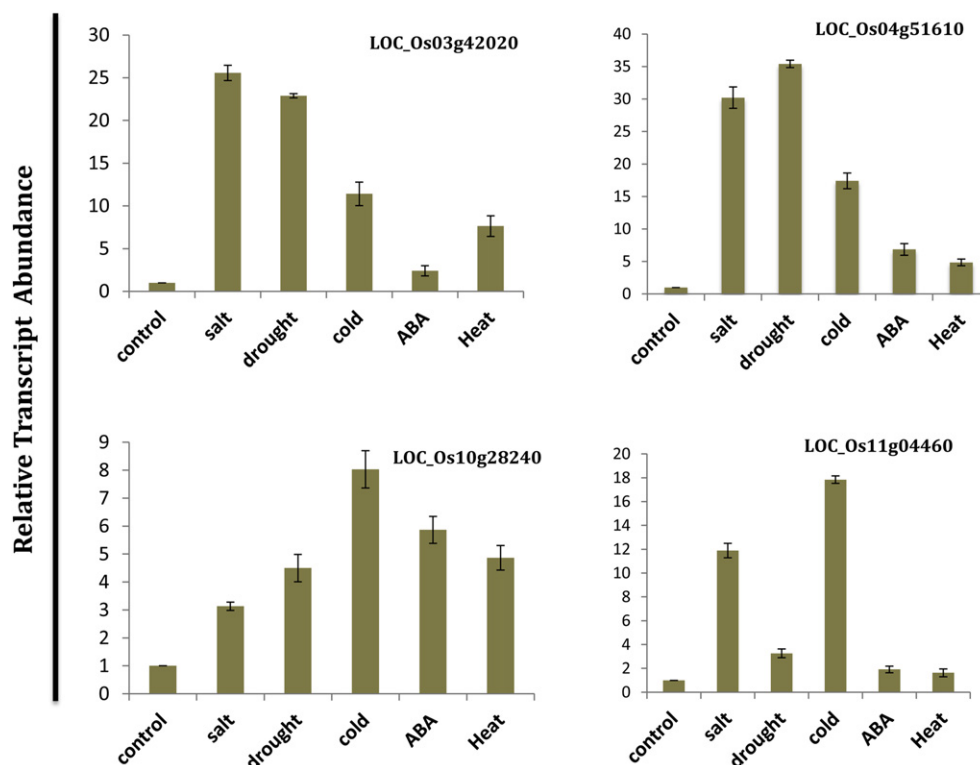


Fig. 7. Relative transcript abundance of selected rice plant-type II Ca^{2+} ATPases using qRT-PCR. Standard error bars have been shown for data obtained from real time PCR. Y-axis represents normalized expression value from real time PCR and X-axis shows different abiotic stress treatment.

Since the localization of the Arabidopsis P-type II Ca^{2+} ATPases has been a predominant criteria for the nomenclature of the gene family and no previous reports on classification and subcellular localization of rice P-type II Ca^{2+} ATPases are available, we have bioinformatically predicted the localization of this gene family in rice. The rice and Arabidopsis type IIA Ca^{2+} ATPases formed a closely associated group in the phylogenetic analysis indicating high sequence similarity which tempted us to predict that probably rice type IIA Ca^{2+} ATPases could have similar functions as its Arabidopsis

counterparts and therefore could also be localized in ER. Subcellular localization revealed that type IIA Ca^{2+} ATPases were present in the PM, tonoplast, ER and Golgi apparatus [21]. A recent proteomic study [22] confirmed the ER localization for Arabidopsis ECA1 and Golgi localization for At-ECA3, but no similar report was found for their rice counterparts. A report on a tomato (*Lycopersicon esculentum*) ECA isoform located in PM suggested that an isoform specific location of every ECA isoform also needs to be experimentally verified [23].

Table 3

Prediction of cis-regulatory elements in plant-type II Ca^{2+} ATPases (rice) promoters with database analysis. ABRERATCAL: response to calcium ion; CGCGBOXAT: involved in multiple signal transduction; IBOXCORE: response to light signal; GT1GMSCAM4: pathogenesis and salt-related; MYB: response to drought and ABA signals; MYC: response to drought, ABA and cold signals; GAREAT: response to GA signal; WBOXNTERF3: response to wound signal; and WBOXATNPR1: response to SA signal.

Types	Element name and number								
Element name	ABRERATCAL	CGCGBOXAT	IBOXCORE	GT1GMSCAM4	MYB	MYC	GAREAT	WBOXNTERF3	WBOXATNPR1
Element core sequence	MACGYGB	VCGCGB	GATAA	GAAAAA	CNGTTR/WAACCA/YAACKG	CANNTG/CATGTG/2CACATG	TAACAAAR	TGACY	TTGAC
Type IIA Ca^{2+} ATPase									
LOC_Os03g17310	6	6	3	5	17	18	0	3	0
LOC_Os03g52090	4	6	1	0	14	12	0	1	0
LOC_Os05g02940	0	0	0	1	17	10	0	4	3
Type IIB Ca^{2+} ATPase									
LOC_Os01g71240	3	4	1	6	4	4	1	2	2
LOC_Os02g08010	0	0	3	0	11	12	2	2	1
LOC_Os02g08018	5	16	0	0	13	2	0	4	0
LOC_Os03g10640	3	10	0	1	7	6	0	4	2
LOC_Os03g42020	1	0	8	5	6	8	0	3	3
LOC_Os04g51610	2	14	2	5	3	10	0	4	2
LOC_Os05g41580	0	0	1	4	4	14	0	1	0
LOC_Os08g40530	0	0	6	2	17	12	1	7	3
LOC_Os10g28240	0	0	1	1	12	4	0	2	2
LOC_Os11g04460	0	2	4	2	12	8	0	2	2
LOC_Os12g04220	0	2	4	1	12	14	0	3	5
LOC_Os12g39660	2	0	6	5	7	14	0	5	3

Yeast *PMR2* showed high similarity to Ca^{2+} ATPases [24] but the genetic evidence indicates that this pump transports sodium [8]. In mammalian type Ca^{2+} ATPases, six amino acid residues are conserved in transmembrane helices M4, M5, M6 and M8, and it has been suggested that they form a high-affinity calcium-binding site [25]. Both type of pumps (ECAs and ACAs) contain 10 transmembrane domains and a large cytoplasmic loop connecting membrane spans 4 and 5. The large central cytoplasmic loop possesses an ATP binding site and the aspartate residue that becomes phosphorylated during the reaction cycle [26].

The type IIA and IIB Ca^{2+} ATPases have similar topology, except that most of the type IIB Ca^{2+} ATPases have an N-terminal autoinhibitory domain while in type IIA this domain is missing. The N-terminal autoinhibitory domain contains a binding site for Ca^{2+} CaM, in addition to a serine phosphorylation site. Binding of CaM to the autoinhibitory site activates the pump activity, while phosphorylation of this domain by CDPK inhibits the pump activity [26]. Animal type IIB Ca^{2+} ATPases are likewise activated by calmodulin, but in this case, the CMBD is situated in an extended C-terminal domain [27]. Their catalytic activity can be modulated by cytosolic $[\text{Ca}^{2+}]$, activation upon binding to CaM, or by inhibition following phosphorylation by Ca^{2+} -dependent protein kinases (CDPKs). Since CaM-binding sites are generally quite diverse, each type IIB Ca^{2+} ATPase may have different affinity for CaM or may bind a different CaM isoform. This domain undergoes reversible sequential phosphorylation inducing conformational changes that may be important for regulating the ATPase function. It has been indicated that C-terminal cation transporting ATPase domains have a regulatory function, either in regulating the activity of the ATPase itself, or in regulating some other cellular function(s).

Nine conserved motifs, characteristic of P-type ATPases superfamily [18], were identified in the present study. The first three motifs are located in the small cytosolic region B-loop present between transmembrane α -helical segment M2 and M3 [2]. The residues within this loop, especially in motif 3, may enhance the stability of this region, thereby providing more favorable reaction kinetics for the transition between the E1 and E2 states. Our results showed that motif 4 (PEGL) is located within and at the end of TMS 5 and probably contributes to energy transduction [2]. Motif 5 (DKTGTLT) is tightly conserved and includes conserved aspartyl phosphorylation site. These residues are well conserved in most P-type ATPases [28]. Motif 6 (KGAPE) is highly conserved in the eukaryotic P-type ATPases [20] and may function in ATP binding [29]. In our study, D (aspartate), the first residue of motif 7 (DPPR), was found to be completely conserved. All functionally characterized and uncharacterized families have the first residue of motif 7 as D [2]. However, the third residues of motif 7 varied substantially. The aspartate (D) residue may influence Mg^{2+} – H_2PO_4^- binding because mutation to N or E results in total or nearly total loss of activity [2]. While the D residue might coordinate the Mg^{2+} , the R residue might be involved in binding the phosphate [29]. Few members of type II Ca^{2+} ATPases showed fully conserved sequence of motif 8 (MVTGD), especially the last three residues (TGD). This motif is involved in the catalysis of enzyme phosphorylation [11]. The function of motif 9 still needs to be validated.

The 3D modeled protein structure of rice and Arabidopsis P-type IIA Ca^{2+} ATPases revealed a transmembrane domain with 10 α -helices. Biochemical evidences also suggested presence of 10 TMS in mammalian type IIA Ca^{2+} ATPases [30]. Structure of the calcium pump from sarcoplasmic reticulum at 8 Å resolution [31] and at 2.6 Å [32] showed two calcium ions bound in the transmembrane domain, which comprises 10 α -helices. The structure of Ca^{2+} pump at 3.1 Å resolution in a Ca^{2+} -free state showed that large-scale rearrangements take place within 6 of the 10 transmembrane helices for the release of calcium ions into the lumen of sarcoplasmic

reticulum [33]. Although we were not able to model the N-terminal autoinhibitory domain due to unavailability of type IIB structural homolog, we have proposed a hypothetical 3D protein structure model with Ca^{2+} binding sites.

To understand evolutionary relationship between different subgroups of P-type II Ca^{2+} ATPases, an unrooted phylogram was constructed using MEGA version 5 with 1000 bootstrap replicates. From the phylogenetic analysis, we found that in rice and Arabidopsis the P-type IIA and IIB Ca^{2+} ATPases have evolved separately from each other. The P-type IIA Ca^{2+} ATPases (ECA pump) from rice form a distinct subset of ER type Ca^{2+} ATPases closely related to Arabidopsis ECA pump proteins. Bootstrap values of the nodes were found to be high in both type IIB and type IIA cladogram. A strong bootstrap value support for individual clades added confidence to the cladogram construction.

The P-type II Ca^{2+} ATPase microarray expression analysis revealed that many of these genes showed differential expression during abiotic stress conditions. The Arabidopsis genes were expressed in all major organs and could be involved in plant adaptation to variable nutrient conditions, playing roles in mineral nutrition [15]. Experiment in yeast suggested that Arabidopsis *ECA1* acts as multi-cation pump that could transport Ca^{2+} , Mn^{2+} and Zn^{2+} [14]. Genetic and biochemical evidence indicated that Ca^{2+} and Mn^{2+} pumped by *ECA1* into ER are critical for growth and tolerance to manganese toxicity [14]. *At-ECA3* is also important for Mn^{2+} and Ca^{2+} homeostasis, possibly functioning in the transport of these ions into the Golgi apparatus. *ECA3* is the first P-type IIA Ca^{2+} ATPase to be identified in plants and is required under Mn^{2+} -deficient conditions [15]. The *Os-ECA1* was first identified as a gibberellic acid-induced gene in the aleurone layer [13]. It therefore seems likely that rice Ca^{2+} ATPases could play an important role in the gibberellic acid signal transduction pathway. The qRT-PCR results of LOC_Os03g42020 (member of type IIA in rice) revealed that the expression of this gene is up-regulated under salt and cold condition as shown in microarray data. However, in case of drought stress condition its expression profile was different. Singh et al. [34] stated that on occasion it is possible for gene expression profiles to be deviated from the anticipated expression pattern. The qRT-PCR analysis also showed slight up-regulation of relative mRNA transcript under heat and ABA stress. Upon exposure to salt stress, enhanced transcript abundance was observed in tomato as well [17]. Similar result was reported by Perez-Prat et al. [18] who observed enhanced Ca^{2+} ATPase gene expression in tobacco culture cells treated with 428 mmol dm^{-3} NaCl. Cytosolic Ca^{2+} changes in response to low temperature and is involved in temperature sensing mechanisms. Cold stress caused a two-fold increase in Ca^{2+} transport activity in plasma membrane in winter rye leaves [35]. This reflected the enhanced capacity of calcium pump to remove Ca^{2+} ions out of the cytoplasm.

Several reports have been published mentioning the fact that type IIB Ca^{2+} ATPases also play important physiological roles. For instance, *At-ACA7* is required for proper pollen development, probably through regulation of Ca^{2+} homeostasis [6], whereas *At-ACA8* is involved in stomatal opening and closing [12]. The *At-ACA9* mutant plants have reduced fertility as pollen tubes fail to grow across the stigma [36]. The *At-ACA10* is another member of ATPase pump, regulating adult vegetative phase development and inflorescence architecture in Arabidopsis [7]. Recent evidence suggests that type IIB Ca^{2+} ATPases (ACAs pump) modulate biotic and abiotic stress response pathways. As shown in the qRT-PCR analysis, these genes revealed up-regulation of their transcripts under different abiotic stresses in rice. The present reports corroborate the result of the microarray analysis generated from the rice array database and emphasize the importance of type IIB Ca^{2+} ATPases gene family in stress signaling pathway. As segmentally duplicated genes bear

same probe set ID in microarray, their individual expression profile cannot be compared. The expression pattern observed in tandemly duplicated genes shows a very least varied pattern which can be considered as similar pattern giving rise to preservation of expression after duplication. In *Physcomitrella patens*, *PCA1* (type IIB Ca^{2+} ATPases) mRNA levels were up-regulated by dehydration, NaCl and abscisic acid. The knockout of *PCA1* resulted into an increased sensitivity to NaCl stress, which was correlated with a NaCl-triggered cytosolic Ca^{2+} elevation, higher in magnitude and longer in duration [16]. Earlier it was also observed that salt treatment increased the relative amount of mRNA in case of *At-ACA4* [3], whereas rapid and dramatic (20-fold) increase was found for *Gm-ACA1* (soybean type IIB Ca^{2+} ATPases) [19]. On the basis of our qRT-PCR result and previous reports, we suggest that both type IIA and IIB Ca^{2+} ATPases might participate in similar pathways when induced under salt stress condition. In tobacco, the *NbCAM1* (type IIB Ca^{2+} ATPases) silencing resulted in an accelerated pathogen-triggered programmed cell death (PCD) response. The authors showed that *NbCAM1* and *NbRBOHB* function downstream of a calcium decoder in immune receptor mediated PCD and proposed that ER type IIB Ca^{2+} ATPases are components of the calcium efflux pathway that controls PCD during an innate immune response [37]. The double knockout mutation of *ACA4* and *ACA11* pumps resulted in a high frequency of hypersensitive response-like lesions [37]. Loss of the vacuolar Ca^{2+} pumps itself does not cause a catastrophic defect in ion homeostasis but rather potentiates the activation of a SA-dependent PCD pathway [5]. These results provided evidence on the activity of vacuolar Ca^{2+} pumps in the control of a SA-dependent PCD pathway in plants.

Genes can be expressed in most plant tissues either by constitutive or inducible-promoters. However, constitutive promoters may result in homology-dependent gene silencing while inducible or tissue specific promoters are required for correct spatial and temporal expression of genes, leading to the regulation of plant development and are preferred as experimental tools to analyze the effects of transgenic expression and produce transgenic plants with resistance to diverse abiotic stresses. PM proteins are involved in the recognition and transduction of endogenous hormonal signals. The present study reveals that P-type II Ca^{2+} ATPase promoter regions in rice harbored multiple-stress cis-acting elements. It has been reported the drought-induced element DRE (dehydration responsive element) usually exists upstream of drought-induced gene promoters, while the ABA-induced gene promoter usually harbors an ABRE (ABA responsive element) [38]. Some genes (e.g. RD22A) contain neither DRE nor ABRE elements, even if they can be induced by both drought and ABA. This kind of gene expression has been reported as regulated by MYB and MYC recognition motifs [39]. The results implied that the P-type II Ca^{2+} ATPases promoters could be induced by drought, salt, cold and ABA through different signal transduction pathways.

In the present study, a comprehensive genome-wide analysis of the P-type II Ca^{2+} ATPase family in rice and Arabidopsis was performed. Although the respective pumps and their genes are remarkably similar, the analysis of the rice sequences has done more than confirming the previous findings in Arabidopsis. We identified 15 members of P-type II Ca^{2+} ATPase in rice and it was further divided into groups on the basis of domain and phylogenetic analysis. An in-depth analysis on gene-expression suggested clear demarcation in roles assigned to each gene. To gain additional knowledge about the P-type II Ca^{2+} ATPase gene family, we searched for previously unreported motifs and checked their importance structurally on the basis of homology modeling. The analysis revealed that the gene family has important roles in diverse functions in plants, both in the developmental stages and in stress. An *in silico* analysis permitted the identification of multiple

environmental and hormonal cis-regulatory elements present in the promoters regions of P-type II Ca^{2+} ATPases in rice. The qRT-PCR results proved their involvement in different abiotic stresses. Potential P-type II Ca^{2+} ATPase genes that are not yet functionally characterized were identified in rice. Once the function of individual members is assigned, the knowledge obtained may allow engineering crop plants with higher productivity and nutritious value by modifying calcium signaling and homeostatic pathways.

4. Methods

4.1. Genome wide search and identification of plant-type II Ca^{2+} ATPases

A keyword based search by name “Plasma membrane Calcium ATPase” was carried out to identify the P-type II Ca^{2+} ATPase genes in the Rice Genome Annotation project (RAP) database (<http://rice.plantbiology.msu.edu/>). The amino acid sequence of candidate genes was checked for presence of characteristic P-type II Ca^{2+} ATPases domains and motifs via S.M.A.R.T. (<http://smart.embl-heidelberg.de>) and Pfam (<http://pfam.sanger.ac.uk/>). Arabidopsis P-type II Ca^{2+} ATPase members were searched based on putative function (Ca^{2+} transporting ATPases) and the sequences were downloaded from Arabidopsis Information Resource (TAIR) database (<http://www.arabidopsis.org/>).

4.2. Sequence similarities, gene nomenclature and attributes

The sequence similarities and divergence chart of all selected P-type II Ca^{2+} ATPase was prepared using MEGALIGN program (DNASTar, Madison, WI). Nomenclature of Arabidopsis Ca^{2+} ATPases was given according to the information of Arabidopsis Information Resource (TAIR) database (<http://www.arabidopsis.org/>). Gene attributes such as locus ID, protein size, introns, alternate spliced forms, evidences in terms of total number of ESTs and full length cDNA were extracted from RGAP version 6.1 and knowledge based *Oryza* molecular biological encyclopedia (KOME) (<http://cdna01.dna.affrc.go.jp/cDNA/>). The localization data for Arabidopsis genes was retrieved from TAIR database and cross-checked with experimentally proven results of more up-to-date SUBA database (<http://suba.plantenergy.uwa.edu.au/>). For rice, subcellular localization was predicted based on the data available in Rice Genome Annotation project (RAP) database (<http://rice.plantbiology.msu.edu/>).

4.3. Protein analysis: sequence, structure and phylogeny

To find out the sequence conservation and functional homology of P-type II Ca^{2+} ATPases in rice and Arabidopsis, multiple sequence alignment of all selected candidate genes was carried out using ClustalX (version 2.0.8) (<http://www.clustal.org/download/2.0.8>). To identify the putative transmembrane domains within P-type II Ca^{2+} ATPases, <http://www.membranetransport.org/> was employed. To search for the conserved regulatory and catalytically important motifs and transmembrane domains the amino acid sequences of all P-type II Ca^{2+} ATPases were aligned and analyzed by employing MegAlign software 5.07 [using ClustalW method].

To infer the evolutionary history of all rice and Arabidopsis P-type II Ca^{2+} ATPases, an un-rooted phylogenetic tree was constructed employing MEGA version using default parameters. The tree was generated by neighbor-joining (NJ) algorithm with p-distance method and gapped pairwise deletion. To test the phylogeny, a bootstrap statistical analysis was performed with 1000 replicates.

The homology models of P-type IIB Ca^{2+} ATPase from rice (LOC_Os04g51610) and Arabidopsis (AT5G57110) were created

using ESyPred3D Web Server 1.0 (<http://www.fundp.ac.be/sciences/biologie/urbm/bioinfo/esypred/>). Molecular graphics visualization program PyMOL was used for visualization and editing of PDB models. The residues of the N-terminal Ca^{2+} binding domain were highlighted.

4.4. Genome organization and gene duplication

The rice P-type II Ca^{2+} ATPase genes were mapped onto chromosomes by identifying their positions as given in RGAP pseudomolecules (version 6.1; chromosome 1–12) according to the coordinates of RGAP loci (<http://rice.plantbiology.msu.edu/>). The RGAP segmental duplication database was explored to find out the duplicated genes (http://rice.plantbiology.msu.edu/segmental_dup/500kb/segdup_500kb.shtml). To visualize P-type II Ca^{2+} ATPase loci in Arabidopsis, chromosomal maps were constructed using the chromosomal map tool available on TAIR database (<http://arabidopsis.org/jsp/ChromosomeMap/tool.jsp>). Genes separated by five or fewer genes were considered as tandem duplications.

4.5. Microarray expression analysis

To get an overview of the expression pattern of rice P-type II Ca^{2+} ATPase genes under various developmental and stress conditions, rice genes were analyzed using Rice Oligonucleotide Array Database (<http://www.ricearray.org/>). Expression values were then fetched to Genesis software v 1.7.5 and a graphical presentation (Heat map) of the data was prepared. The clustering of genes was done by hierarchical clustering using average linkage clustering as rule. The data of Arabidopsis expression analysis carried out by using Affymetrix 22K. ATH1 chip was obtained from Genevestigator response viewer (<https://www.genevestigator.com>).

4.6. Quantitative expression analysis by real time PCR

For transcript analysis, rice seedlings (*Oryza sativa* ssp. *indica* var. IR64) were grown in greenhouse under normal conditions and subjected to various stress conditions. For abiotic stress treatments, seedlings were transferred to salt solutions (200 mM NaCl) or abscisic acid (100 μM ABA) at room temperature for 3 h. For cold and heat treatments, the plantlets were transferred to incubators at 4 °C and 42 °C, respectively. To mimic drought stress, seedlings were air dried on a Whatman sheet for 3 h. Parallel control samples were prepared by keeping the seedlings in water for 3 h. Treated seedlings were immediately frozen in liquid nitrogen. The RNA of the stressed samples was isolated as previously described and cDNA was prepared using AccuScript™ High Fidelity 1st Strand cDNA Synthesis kit (Agilent Technologies, USA) as per manufacturer's instructions.

Real time PCR was carried out to validate the microarray data for a few selected genes, showing significant differential expression pattern under abiotic stress conditions. The primers were designed for all the selected genes using PRIMER EXPRESS (PE Applied Biosystems, USA) using default parameters. Primers were checked using BLAST tool of NCBI and dissociation curve was analyzed after the PCR reaction for their specificity. List of primers used for qRT-PCR are given in (Supplementary Table 1). The reaction was set up using Brilliant III Ultra Fast SYBR Green QPCR Master Mix (Agilent Technologies, USA) in ABI 7500 Real Time PCR System (Applied Biosystems, USA). The cDNA variance among samples was normalized using α -tubulin as the endogenous control. Relative expression values were calculated by $\Delta\Delta\text{Ct}$ method. Biological and technical replicates of each sample were taken for the analysis.

4.7. Analysis of cis-elements in rice plant-type II Ca^{2+} ATPases promoter regions

The 1-kb upstream sequences of rice P-type II Ca^{2+} ATPase genes were downloaded from the data available in Rice Genome Annotation project (RAP) database (<http://rice.plantbiology.msu.edu/>). Promoter elements for all the sequences were analyzed by PLACE (<http://www.dna.affrc.go.jp/PLACE/>) and PLANTCARE (<http://bioinformatics.psb.ugent.be/webtools/plantcare/html/>).

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.plaphy.2013.01.002>.

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